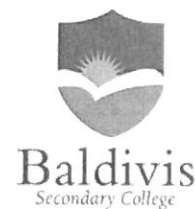


Year 8 Mathematics
Mini Test
Numbers and Indices



Resources: one A4 page of notes

Name: Sol

/ 23 marks

Time: 20 mins

Full working out must be shown to get full marks. Attempt all the questions.

Part A – Directed Numbers

10 marks

Evaluate:

1) $-2 + 7$

$= 5$ ✓

2) $9 + (-3) =$

6 ✓

3) $-17 - 3 =$

-20 ✓

4) $2 - (-10) =$

12 ✓

5) $-5 \times (-6) =$

30 ✓

6) $-24 \div 3 =$

-8 ✓

7) $5 + 2 \times (-3) =$

$5 + -6$ ✓
 $= -1$ ✓

8) $2 \times (-7) - (-12) \div (-4)$

$-14 - 3$ ✓
 $= -17$ ✓

Part B – Index Laws**13 marks**

1) Write the following in expanded form

a) $4^2 = 4 \times 4$ ✓

b) $2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2$ ✓

2) Simplify and leave the answer in index form. (1, 1, 1, 1, 2 marks)

a) $3^5 \times 3^4 = 3^9$ ✓

b) $5^9 \div 5^6 = 5^3$ ✓

c) $7 \times 7^2 \times 7^3 \times 8^3 = 7^6 \times 8^3$ ✓

d) $(9^5)^3 = 9^{15}$ ✓

e) $2 \times 4^0 \times 2^7 \times (2^2)^3 = 2^{14}$ ✓✓

3) Evaluate the following. (1, 1, 1, 2)

a) $17^0 = 1$ ✓

b) $10 \times 6^0 = 10$ ✓

c) $3^2 \times 2 \times 5^0 = 9 \times 2 \times 1 = 18$ ✓

d) $\frac{(6^2)^4}{6^5 \times 6^2} = \frac{6^8}{6^7} = 6^1 = 6$ ✓

End of Test